Species-Specific Responses of Urban Tree Phenology to Climate Drivers in a Semi-Arid Environment

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Abstract

Urban trees provide critical cooling benefits in cities, but their effectiveness depends on phenological timing—the seasonal patterns of leaf emergence, growth, and senescence. As climate change intensifies urban heat islands, understanding how tree phenology responds to temperature and precipitation drivers becomes essential for climate adaptation planning. This study leverages near-daily PlanetScope satellite observations (2018-2024), climate data, and land cover mapping to investigate phenological responses of 12 common deciduous tree species in Denver, Colorado. We extracted four phenological stage dates (greenup, maturity, senescence, dormancy) for 3,450 individual trees using automated curve-fitting methods, then modeled phenological timing against seasonal climate variables and land cover characteristics using XGBoost machine learning. Climate responses varied substantially by phenological stage: spring temperature strongly controlled maturity timing across all species, while senescence responded to both late-summer temperature and precipitation drivers, and greenup and dormancy showed weaker, more complex climate associations. Partial dependence analysis revealed important non-linear responses, including a 32°C threshold where heat stress accelerates senescence and a maturity plateau at 23°C where thermal acceleration saturates. Land cover effects were modest compared to climate drivers, with irrigation showing clearer effects than impervious surfaces. Climate warming scenarios project substantial phenological shifts: under observed +1.1°C warming over the past 30 years, growing seasons have extended by an average of 9.1 days, driven primarily by delayed dormancy rather than earlier greenup. Species showed divergent responses, with highly plastic species (Tree of Heaven, Western Catalpa, Common Hackberry) extending growing seasons by 15-16 days, while conservative species (cottonwoods, Northern Red Oak) showed minimal changes. These findings have important implications for urban forest management and ecosystem service provision under continued climate change.

1. Introduction

Urban trees are among the most effective natural mitigators of the urban heat island (UHI) effect. Studies have demonstrated that neighborhoods with mature tree canopy cover experience measurably cooler microclimates compared to areas with limited vegetation (Ziter et al. 2019; Alonzo et al. 2021). As global temperatures continue to rise, the effects of UHIs are expected to intensify in the coming decades (Stone 2012). Given the increasingly critical role of urban forests in climate resilience strategies, it is essential to understand the drivers of urban tree phenology—the seasonal patterns of leaf emergence, growth, and senescence—and how these patterns may shift as environmental conditions change.

The timing and duration of these seasonal patterns are strongly influenced by climate, particularly temperature and precipitation. Elevated temperatures caused by climate change can lengthen the growing season of urban trees (Zhang et al. 2022; Yang et al. 2020). Higher spring temperatures typically accelerate leaf emergence, while warmer fall temperatures may delay leaf senescence, extending the period of active growth and canopy cover (Zhang et al. 2022; Zhou et al. 2020). This extended canopy period can enhance the cooling benefits of urban trees by increasing shade and evapotranspiration, helping to mitigate the urban heat island effect. However, these benefits are not guaranteed: some studies have shown that warmer winters followed by late spring freezes can delay leaf-out, disrupt flowering, and reduce the expected benefits of extended growing seasons, particularly in species that leaf out early in spring (Chamberlain et al. 2021; Fu et al. 2024).

Precipitation patterns add another layer of complexity. Spring drought can delay leaf-out and flowering, while wet conditions may accelerate growth but increase vulnerability to disease and root stress. In the fall, low precipitation or drought stress can trigger earlier leaf senescence, shortening the period of active growth and reducing canopy cover (Cleland et al. 2007; Peñuelas et al. 2004; Li et al. 2019). Critically, the interaction of temperature and precipitation determines ultimate outcomes, as insufficient soil moisture can counteract the effects of warmer temperatures, underscoring the complex challenges that climate change poses for urban tree health and their role in mitigating urban heat.

Monitoring and predicting these phenological responses require appropriate observational tools. The body of research on urban phenology has evolved along multiple scales of observation. Coarse-scale studies often rely on remote sensing platforms such as MODIS, which provide high temporal but low spatial resolution data to monitor citywide phenological trends and seasonal canopy dynamics (Zhang et al. 2004; Melaas et al. 2016). At the opposite end of the spectrum, fine-scale studies use in-situ observations or near-surface imaging networks like PhenoCam to capture species-level variation and microclimate effects that are often obscured at broader scales (Richardson et al. 2018; Klosterman et al. 2018). More recently, high-resolution satellite constellations such as PlanetScope have emerged as a valuable intermediate approach, combining the spatial precision needed to monitor the location of individually identified trees with the temporal frequency required to monitor phenological changes at an almost daily basis (Bolton et al. (2020); Alonzo et al. (2021)).

Researchers can now study urban vegetation across large urban areas on an unprecedented detail and scale.

1.1 Objectives

This study leverages near-daily PlanetScope NDVI observations of urban trees in Denver, Colorado, daily climate data from 2018 to 2024, and land cover data to investigate how urban tree phenology responds to environmental drivers and how these responses may shift under future climate scenarios. Specifically, this research addresses three key questions:

1. Does temperature or precipitation more strongly influence the timing of greenup, maturity, senescence, and dormancy for twelve common urban tree species in Denver, and are these responses fairly uniform across species?
2. How does land cover—such as impervious surfaces versus irrigated landscapes—affect tree phenology?
3. How have/will different tree species respond to moderate climate change scenarios?

1.2 Study Area

Denver, Colorado provides a compelling case study for understanding urban tree phenology. Located in the mid-latitudes, the city experiences distinct seasonal patterns that produce clear phenological transitions across all four stages (greenup, maturity, senescence, and dormancy). The city maintains a detailed public tree inventory representing hundreds of species, and detailed land cover mapping enables analysis of how urban infrastructure affects these phenological patterns.

As the planet warms, Denver is projected to confront both warming and precipitation uncertainty—conditions that will directly affect the timing of leaf-out, canopy development, and senescence for this urban forest. According to Colorado State University's Colorado Climate Center's most recent Climate Change in Colorado report, by mid-century, Denver is projected to see temperatures rise between 1.4°C and 3.1°C compared to the 1971–2000 baseline. The city has already warmed by approximately 1.1°C over the past 30 years, with the most pronounced increases occurring during summer and fall. Under moderate emissions scenarios, Denver's climate will resemble that of Pueblo, Colorado today, whereas higher emissions scenarios could make it feel more like Albuquerque, New Mexico (Bolinger et al. 2024).

Summer and fall are expected to warm slightly more than winter and spring, with summer temperatures increasing by 2.2–3.3°C and winter by 1.7–2.2°C. This warming is projected to drive dramatic increases in extreme heat events: by mid-century, Denver could experience an average of 7 days per year exceeding 37.8°C (100°F) and roughly 35 days above 35°C (95°F), compared to very few such days historically. In essence, the typical year in 2050 may be as warm as the hottest years on record today (Bolinger et al. 2024).

Future precipitation patterns for Denver remain uncertain. Climate models (CMIP5 and CMIP6 under RCP 4.5 scenarios) disagree on whether precipitation will increase or decrease, with projections ranging from -7% to +7% by 2050. Even with potential increase in precipitation, warmer temperatures may offset benefits by pulling more moisture out of the soils and melting snowpacks reducing water availability for trees (Bolinger et al. 2024).

The study period of this paper (2018 to 2024), captures years with temperatures and precipitation both above and below the 1971-2000 baseline. Notably, 2024 was Denver's third warmest year on record, with temperatures 2°C above baseline—approximating mid-century projections—while 2019 and 2020 were notably drier than average. This climate variability enables us to model how Denver's urban trees respond to average conditions projected for 2050 (National Oceanic and Atmospheric Administration 2024).

2. Methods & Materials

This study models individual tree phenological metrics against corresponding land cover characteristics and seasonal climate variables using machine learning methods. We combined near-daily PlanetScope NDVI observations from 2018-2024 with Denver's municipal tree inventory, daily climate records, and high-resolution land cover data to create a comprehensive dataset linking individual tree phenology to environmental drivers. XGBoost models were fit separately for each species and phenological stage (greenup, maturity, senescence, dormancy) to quantify climate and land cover effects, with model predictions then used to simulate phenological responses under future climate scenarios.

2.1 Materials

2.1.1 Denver Tree Inventory and Tree Sample Selection

The City and County of Denver's *Tree Inventory Denver* is a publicly available urban forestry database containing detailed records for approximately 300,000 trees across Denver County. The inventory includes species identification, tree size measurements, health assessments (disease and pest observations), and precise GPS coordinates for each tree. This comprehensive dataset enabled our remote sensing analysis by allowing us to link individual trees in the inventory to their corresponding locations in satellite imagery (City and County of Denver Department of Parks and Recreation, n.d.).

Our analysis focused exclusively on trees within Denver County boundaries west of longitude -104.7302°. This spatial filter excluded outlier trees that, while technically within county boundaries, exist in radically different environments such as prairie grasslands near the airport.

From this spatially filtered inventory, we implemented a four-part sampling approach to select trees suitable for satellite-based remote sensing phenology analysis. First, we restricted the sample to mature trees with diameters of 24 inches or greater, as larger trees have crown sizes more readily detectable in moderate-resolution satellite imagery. Second, we selected 15 species based on their abundance in the Denver urban forest, ensuring adequate sample sizes for species-specific modeling. Third, we prioritized species diversity by including both deciduous and coniferous species, representatives from multiple genera with intra-genus variation (e.g., multiple *Populus* species), and a mix of native and non-native species to capture variation in climate sensitivity and adaptation. Fourth, we considered average mature crown size for each species based on USDA Forest Service Urban Tree Database classifications to ensure trees would be adequately resolved in PlanetScope imagery (McPherson et al. 2016).

An interactive Shiny application was developed to facilitate sample selection and quality control, allowing visual inspection of candidate trees' spatial distribution across Denver County. The final sample consisted of 23,779 individual trees from 15 species representing 11 genera, including 11 native species, 2 non-native species, and 2 invasive species (Table 1).

Table 1: Summary of selected tree species with sample sizes and ecological characteristics

Scientific Name	Common Name	Genus	Obs.	Status	Leaf Type
Acer saccharinum	Silver Maple	*Acer*	6,217	Native	Deciduous
Ailanthus altissima	Tree of Heaven	*Ailanthus*	426	Invasive	Deciduous
Catalpa speciosa	Western Catalpa	*Catalpa*	656	Native	Deciduous
Celtis occidentalis	Common Hackberry	*Celtis*	656	Native	Deciduous
Fraxinus pennsylvanica	Green Ash	*Fraxinus*	2,369	Native	Deciduous
Gleditsia triacanthos	Honey Locust	*Gleditsia*	2,642	Native	Deciduous
Pinus nigra	Austrian Pine	*Pinus*	957	Non-native	Conifer
Pinus ponderosa	Ponderosa Pine	*Pinus*	521	Native	Conifer
Populus deltoides	Eastern Cottonwood	*Populus*	871	Native	Deciduous
Populus sargentii	Plains Cottonwood	*Populus*	1,070	Native	Deciduous
Quercus rubra	Northern Red Oak	*Quercus*	481	Native	Deciduous

Scientific Name	Common Name	Genus	Obs.	Status	Leaf Type
Tilia americana	American Linden	*Tilia*	655	Native	Deciduous
Tilia cordata	Littleleaf Linden	*Tilia*	405	Non-native	Deciduous
Ulmus americana	American Elm	*Ulmus*	2,794	Native	Deciduous
Ulmus pumila	Siberian Elm	*Ulmus*	2,919	Invasive	Deciduous

2.1.2 Remotely Sensed NDVI Data

The Normalized Difference Vegetation Index (NDVI) is a widely used spectral index that measures vegetation greenness by comparing the difference between near-infrared (strongly reflected by vegetation) and red light (absorbed by vegetation). This relationship emerges from plant physiology: chlorophyll in leaves strongly absorbs red light for photosynthesis, while the cellular structure of healthy leaves strongly reflects near-infrared radiation. The index ranges from -1 to +1, with higher values indicating greater photosynthetic activity and canopy density (Rouse et al. 1973; Tucker 1979).

Since its development in the 1970s, NDVI has become one of the most widely applied metrics in vegetation remote sensing. It has been used to monitor crop health, assess drought impacts, track deforestation, and map global vegetation dynamics across scales from individual plants to entire biomes. For phenological applications specifically, NDVI provides a continuous, objective measure of canopy greenness that directly reflects the seasonal cycles of leaf emergence, maturation, and senescence.

The studies NDVI observations were obtained from PlanetScope's imagery in the red (590-682 nm) and near-infrared (780-888 nm) bands (Planet Labs PBC 2024).

The PlanetScope imagery was then processed using methodology used in Alonzo et al. (2023). The fine resolution, multiple daily images had to be mosaicked togeather to create continuous spatial coverage over Denver County, with scene overlaps resolved by selecting the highest quality pixels based on image acquisition geometry and radiometric quality. Quality assessment layers provided by PlanetScope were used to mask clouds, snow, and haze, automatically detecting atmospheric interference that would compromise NDVI accuracy. NDVI was calculated for each masked image using the standard formula:

$$NDVI = (NIR - Red)/(NIR + Red)$$

Tree-level NDVI values were extracted by overlaying tree locations from the Denver inventory with NDVI imagery. For each tree, we created a circular buffer around the GPS coordinates sized to approximate the tree crown based on species-specific mature crown diameter estimates. To capture canopy pixels rather than understory or adjacent features, we filtered each tree-date observation to retain only the top 25% of NDVI values within the buffer, reducing contamination from grass, pavement, or other non-target features while maintaining sufficient sample size. These daily observations were compiled into NDVI time series for each tree spanning 2018-2024, creating a multi-year record of vegetation greenness dynamics at the individual tree level that served as the foundation for phenological curve fitting and metric extraction (Figure 1).

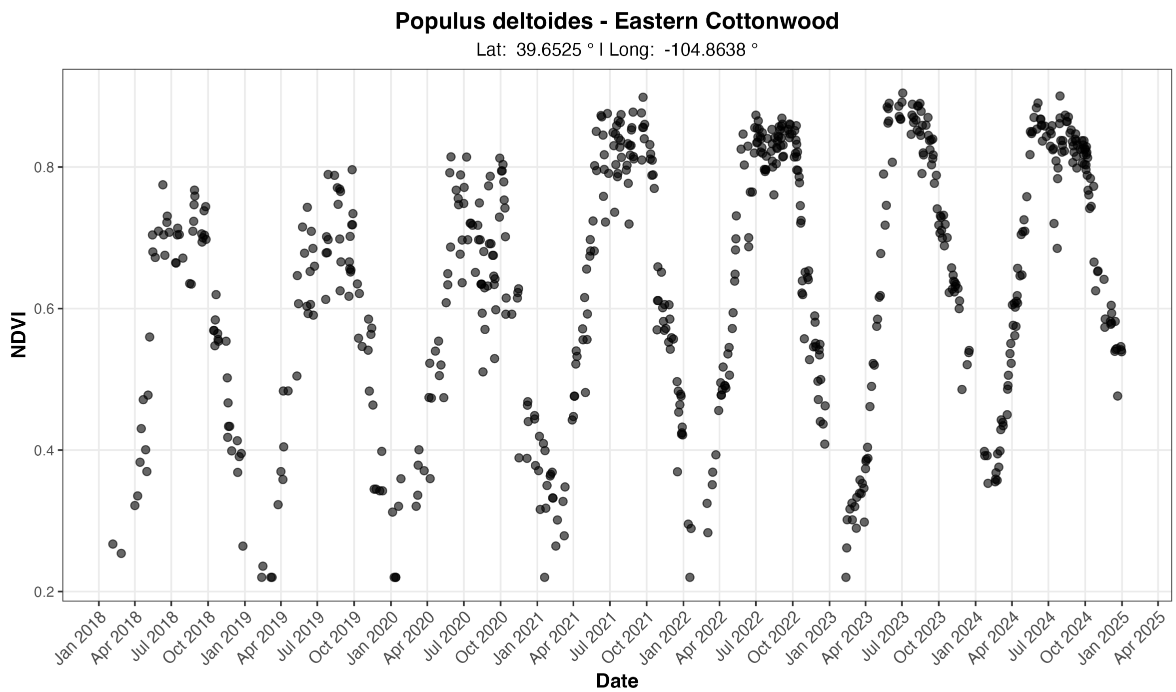

Figure 1: Example multi-year NDVI time series for one tree from our sample of 23,779 trees.

2.1.3 Extracted Phenological Metrics

Phenological metrics were extracted from the raw NDVI time series using the `phenofit` package in R (Kong et al. 2022), which implements multiple curve-fitting methods and automated quality control procedures for vegetation phenology analysis. Raw NDVI values were first despiked to remove anomalous observations caused by residual cloud contamination, sensor noise, or atmospheric interference using a median-based despike filter with a 3-point window and 5 standard deviation threshold from the `oce` package (Kelley and Richards 2022).

Individual growing seasons were identified from the NDVI time series using the `season_mov()` function with weighted Whittaker (wWHIT) smoothing (Eilers 2003; Kong et al. 2019), which automatically detects seasonal boundaries by identifying local minima in smoothed NDVI curves. To improve the accuracy of growing season detection, particularly for trees exhibiting multi-year growth trends, NDVI time series were detrended using a linear model regressing NDVI against time. The fitted trend was removed while preserving the mean NDVI value, allowing the season detection algorithm to isolate seasonal phenological cycles from inter-annual growth patterns. While growing seasons generally corresponded to calendar years, the algorithm occasionally detected seasons spanning year boundaries, particularly for trees with extended dormancy periods or delayed spring greenup. For each detected growing season, we fitted phenological curves using the double-logistic Elmore function (Elmore et al. 2012), which models vegetation greenness as

$$y(t) = c + (d - c)\left[\frac{1}{1 + e^{a_1(t-b_1)}} - \frac{1}{1 + e^{a_2(t-b_2)}}\right]$$

where c is the baseline NDVI, d is the peak NDVI, (a_1, b_1) control spring greenup rate and timing, and (a_2, b_2) control fall senescence rate and timing. The Elmore curve was selected after testing multiple curve types (double-logistic, asymmetric Gaussian, Beck) because it consistently produced the most reliable fits across species and provided biologically interpretable parameters for both greenup and senescence phases.

Curve fit quality was assessed using adjusted R^2, with trees retained in the final sample only if they met two criteria: (1) mean R^2 0.90 across all fitted growing seasons, and (2) at least 6 growing seasons successfully detected over the 7-year study period. These conservative thresholds ensured high-quality phenological curves while excluding trees with poor NDVI signal due to crown damage, sensor artifacts, or persistent cloud cover. Four phenological stage dates were extracted from each annual curve using the inflection point method (Zhang et al. 2003), which identifies the dates of maximum rate of change in the NDVI trajectory: (1) Greenup - spring inflection point marking rapid leaf emergence, (2) Maturity - the date of peak NDVI when maximum canopy development is reached, (3) Senescence - fall inflection point marking rapid leaf loss, and (4) Dormancy - the date when NDVI reaches its winter minimum. Inflection points were chosen over alternative methods (e.g., threshold-based or derivative-based) because they are less sensitive to baseline NDVI variations and provide consistent timing estimates across species with different canopy densities. All extracted metrics, fitted curve parameters, goodness-of-fit statistics, and curve metadata were saved for each tree-year observation. A custom R function was developed to facilitate visualization of annual curves with overlaid phenological metrics, available at [GitHub link]. After quality filtering, 3,450 trees across 12 species remained from the initial sample of 23,779 trees, providing 24,150 tree-year observations across the 2018-2024 study period. The distribution of tree species in the final dataset is shown in Table X. The final dataset consisted entirely of deciduous species, as coniferous trees did not pass the quality filtering checks.

2.1.4 Climate Data

Daily climate observations were obtained from the National Oceanic and Atmospheric Administration (NOAA) Global Historical Climatology Network-Daily (GHCN-Daily) database for station USW00023062, located within Denver Central Park (formerly Stapleton International Airport). This station was selected due to its central location within Denver County. Three core meteorological variables were extracted: maximum daily temperature (TMAX), minimum daily temperature (TMIN), and daily precipitation (PRCP). From these variables, Daily mean temperature was calculated as the arithmetic mean of TMAX and TMIN (National Oceanic and Atmospheric Administration 2024).

To align climate variables with the biological timing of phenological stages, daily observations were aggregated into seasonal summaries corresponding to key periods of tree development (Dixon et al. 2025). Temperature metrics included both seasonal mean temperatures and seasonal mean maximum temperatures, capturing both average thermal conditions and peak daytime heat stress. Precipitation was summarized in two ways: seasonal totals (e.g., spring precipitation sum) to capture resource availability during active growth periods, and year-to-date cumulative totals to represent antecedent moisture conditions that influence phenological timing through soil water storage. These seasonal windows were defined to capture climate conditions during biologically relevant periods preceding or coinciding with each phenological stage. For example, greenup timing was modeled using late winter and early spring climate (February-April), while senescence timing incorporated summer and early fall conditions (June-October).

The resulting climate dataset included 10 distinct seasonal periods (winter, late winter, spring, early spring, late spring, summer, late summer, fall, early fall, late fall) with corresponding temperature and precipitation metrics. This temporal structure was designed to maximize consistency across phenological stages while preserving sufficient climate variation to detect species-specific sensitivities to temperature versus precipitation drivers. Table 2 provides a complete breakdown of seasonal climate variables, their corresponding months, and which phenological were selected as predictors for modeling.

Table 2: Climate Variable Definitions and Seasonal Aggregations

Seasonal Period	Months	Climate Metrics	Rationale
Winter	Dec-Jan-Feb*	Mean temp, Mean Tmax, Total precip, Precip days	Winter chilling requirements and moisture storage
Late Winter	Feb-Mar	Mean temp^{G}, Mean Tmax^{G}, Total precip^{G}, Precip days^{G}, YTD cumulative precip	Transition to spring; pre-greenup conditions

Seasonal Period	Months	Climate Metrics	Rationale
Spring	Mar-Apr-May	Mean temp, Mean Tmax, Total precip[M], Precip days, YTD cumulative precip	Primary growing season onset
Early Spring	Mar-Apr	Mean temp, Mean Tmax, Total precip, Precip days, YTD cumulative precip	Critical leaf-out period
Late Spring	May-Jun	Mean temp[M], Mean Tmax[M], Total precip, Precip days[M], YTD cumulative precip	Canopy development and peak growth
Summer	Jun-Jul-Aug	Mean temp, Mean Tmax, Total precip, Precip days, YTD cumulative precip	Peak growing season and heat stress
Late Summer	Jul-Aug	Mean temp, Mean Tmax, Total precip[S], Precip days[S], YTD cumulative precip	Transition to dormancy preparation
Fall	Sep-Oct-Nov	Mean temp, Mean Tmax[D], Total precip, Precip days[D], YTD cumulative precip[D]	Leaf senescence period
Early Fall	Sep-Oct	Mean temp[S], Mean Tmax[S], Total precip, Precip days, YTD cumulative precip	Primary leaf color change and drop
Late Fall	Nov-Dec	Mean temp[D], Mean Tmax, Total precip, Precip days, YTD cumulative precip	Final leaf drop and winter preparation

[a] December values assigned to following year to maintain biological relevance
[b] Used in Greenup models
[c] Used in Maturity models
[d] Used in Senescence models
[e] Used in Dormancy models

Note: All models also included land cover variables (impervious surface, irrigated surface, unirrigated surface, tree canopy, and water at 90m buffer) which are detailed in Section 2.1.5.

2.1.5 Landcover Data

Land cover characteristics surrounding each tree were derived from the City and County of Denver's high-resolution land cover dataset, which provides 1-meter spatial resolution classification of the urban landscape. This dataset classifies the region into nine distinct land cover categories: (1) structures, (2) impervious surfaces, (3) water, (4) grassland/prairie, (5) tree canopy, (6) irrigated lands/turf, (7) barren/rock, (8) cropland, and (9) shrubland/scrubland (Colorado Information Marketplace, n.d.). The fine spatial resolution of this dataset allows for precise characterization of the immediate environment surrounding individual trees, capturing the heterogeneity of urban landscapes that influences tree microclimate and resource availability.

To quantify local land cover context for each tree, circular buffers of 90 meters were created around each tree's GPS coordinates. These buffer sizes were selected to capture both immediate microsite conditions on tree phenology. Within each buffer, the number of pixels belonging to each land cover class was tallied and converted to percentages, representing the proportion of the surrounding landscape occupied by each cover type. Following methodology similar to (Crawford et al. 2024), the nine original land cover classes were consolidated into five ecologically meaningful categories to simplify interpretation and reduce model complexity (Table 3). The 90-meter buffer land cover variables were used as predictors in all XGBoost models to account for local environmental context effects on phenological timing.

Table 3: Land Cover Classification Consolidation

Final Category	Original Classes Included
Impervious Surface	Structures, Impervious Surfaces, Barren/Rock
Irrigated Surface	Cropland, Irrigated Lands/Turf
Unirrigated Surface	Grassland/Prairie, Shrubland/Scrubland
Tree Canopy	Tree Canopy
Water	Water

2.2 Methedologies

2.2.1 Pearson Correlation Analysis

Prior to formal modeling, we conducted exploratory correlation analysis to assess the strength and direction of relationships between climate variables and phenological metrics across species. We calculated Pearson correlation coefficients between each phenological metric and its biologically relevant climate predictors using pairwise complete observations to handle missing data. This approach allowed us to examine preliminary patterns while accommodating the unbalanced nature of our dataset.

For each species, we computed correlations between four phenological metrics (green-up, maturity, senescence, and dormancy) and their corresponding climate variables selected based on seasonal relevance. This dual approach allowed us to assess both the consistency of climate effects across species and the degree of species-specific variation in climate sensitivity. Results are visualized as a heatmap (Figure 2) showing individual species correlations alongside the cross-species average.

2.2.2 Extreme Gradient Boosted Model

We used XGBoost (Extreme Gradient Boosting) to model the relationship between environmental predictors and phenological timing for each tree species. XGBoost is a machine learning algorithm that builds an ensemble of decision trees sequentially, where each new tree learns to correct the errors of previous trees. Unlike traditional parametric models, XGBoost makes no distributional assumptions about residuals and can automatically capture non-linear relationships and complex interactions between predictors. Individual models were fitted for all four of the phenological metrics for each of the 12 species (48 models in total).

For each if model, XGBoost learns a function where:

$$Y = f(X_1, X_2, ..., X_k)$$

with Y representing the phenological metric (day of year), $X_1 ... X_k$ representing environmental predictors (climate and land cover variables), and tree features capturing individual tree characteristics. The function $f()$ is learned through an ensemble of regression trees, with each tree making sequential improvements to minimize prediction error.

Each model included two categories of features. First, we incorporated environmental predictors specific to each phonological stage, including 3 climate variables (seasonally relevant temperature mean, mean max temperature and precipitation either year-to-date cumulative or cumulative within a given season) and land cover variables (percent impervious surface and percent of irrigated surfaces). Climate variables captured regional meteorological conditions and year-to-year variation, while land cover variables represented local environmental context including urban heat island effects, altered water availability, and differences in solar radiation exposure. When both temperature and precipitation variables were included for a phenological stage, we also created an interaction term (temperature × precipitation) to allow the model to capture synergistic or antagonistic effects between these climate drivers. For instance, the phenological response to warm temperatures may differ under wet versus dry conditions. Below is a breakdown of the climate and land cover variables used to models each of the phonological metrics.

Second, we included tree-specific features to account for repeated measures on individual trees, analogous to random effects in mixed models. These features included tree_mean_pheno (historical average phenology timing for each tree), tree_n_obs (number of observation years per tree), and UID_numeric (numeric tree identifier). These tree-specific features allow the model to capture consistent early or late timing in individual trees due to genetics, tree health, or fine-scale microsite conditions, while environmental predictors capture spatial and temporal variation.

We used conservative hyperparameters optimized for regularization and generalization. The ensemble consisted of 500 trees (nrounds), with each tree having a maximum depth of 4 levels (max_depth) to prevent overfitting. The learning rate (eta) was set to 0.01, a relatively low value that requires more iterations but produces more stable and accurate models. To introduce beneficial randomness that improves generalization, we set subsample to 0.7 (using 70% of observations for each tree) and colsample_bytree to 0.7 (using 70% of features for each tree). The objective function was set to reg:squarederror for squared error loss appropriate for continuous outcomes. These conservative parameters prioritize prediction accuracy and model stability over training speed.

We evaluated model performance using 10-fold cross-validation with a fixed random seed (123) for reproducibility. Data were randomly partitioned into 10 groups, with models iteratively trained on 90% of data and tested on the remaining 10%. This procedure was repeated across all folds to produce out-of-sample predictions for the entire dataset. We calculated four performance metrics: cross-validated R^2 (variance explained), root mean squared error (RMSE, in days), and mean absolute error (MAE, in days). The average performance of all the species models by phenological stages can be found in Table 4.

Our models achieved RMSE values ranging from 5.9 to 12.9 days across all 48 species-stage combinations. These error magnitudes are approximately 2-3 times higher than RMSE values (2.1 to 4.9 days) reported by (Dai et al. 2019) using gradient boosting for the prediction of tree phenology metrics (spring leaf-out). The significant difference could be attributed to differences in data sources used to derive the phenological metrics. (Dai et al. 2019) modeled spring leaf-out dates from ground-based visual observations at individual sites over a 20 to 30 year period, instead of remote sensing data collected over a 7 year period. Given the limitations of our data a our gradient boosting models are preforming within an exceptionable margin of error and are effectively capture climate-phenology relationships.

Table 4: XGBoost Model Performance by Phenological Stage

Phenological Stage	N Species	Total Obs.	Mean R^2	Mean RMSE (days)	Mean MAE (days)
Maturity	12	2,244	0.644	6.95	5.11
Green-up	12	2,194	0.644	8.29	6.32
Senescence	12	2,182	0.691	9.10	7.01
Dormancy	12	2,158	0.590	11.64	8.87

A complete breakdown of performance across all 48 individual species-stage models (Table 5), as well as a breakdown of the importance of each variable in the models (**?@tbl-importance**) can be found in Appendix . All analyses were conducted in R version 4.x (R Core Team 2024) using the xgboost (Chen et al. 2024), lme4 (Bates et al. 2015), mgcv (Wood 2017), caret (Kuhn 2024), and tidyverse (Wickham et al. 2019) packages.

2.2.3 Linear Mixed Models: Comparison and Limitations

We initially attempted to model climate-phenology relationships using linear mixed-effects models (LMMs), which are standard in phenological research due to their interpretability. LMMs were specified as

$$Y_{ij} = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + u_i + \varepsilon_{ij}$$

where Y_{ij} is phenology timing for tree i in year j, X_1 and X_2 are climate predictors, β_0-β_2 are fixed effect coefficients, u_i is a random intercept for each tree, and ε_{ij} is residual error.

However, diagnostic tests revealed severe assumption violations across 97% of models (69/72). Only 2.8% (2/72) passed tests for normality, homoscedasticity, and random effect normality. The mean proportion of outliers ($|$standardized residual$| > 3$) was X.X%, approximately 2-3× higher than expected under perfect normality. Visual diagnostics showed heavy-tailed residual distributions and heteroscedasticity, raising concerns about the reliability of standard errors and coefficient estimates despite large sample sizes (336-5,359 observations per model).

While large sample sizes and cross-validation suggested LMMs remained reasonable approximations, the pervasive assumption violations raised concerns about parameter estimate reliability and standard error accuracy. We tested Generalized Additive Mixed Models (GAMMs) as an alternative that relaxes normality assumptions, but GAMMs either performed worse than LMMs or failed to converge due to limited basis dimension relative to the number of trees. These limitations motivated our adoption of XGBoost as the primary modeling framework.

3. Results

3.1 Species-Specific Climate Sensitivities

3.1.1 Climate-Phenology Correlations

Pearson correlation analysis revealed distinct patterns of climate sensitivity across phenological stages and species. Temperature and precipitation exert differential influence depending on the seasonal timing of each phenological transition.

Although the relationship between climate metrics and greening up timing is weaker than the other three phenological stages, across all species, greening up timing seemed to be driven most notably by late winter climate conditions. Late winter precipitation metrics show the strongest negative correlations with greening up timing. This suggests that wetter late winter conditions advance spring leaf emergence, having a greater effect than temperature during this period. Notably, the American Elm, Siberian Elm and Silver Maple exhibited slightly stronger sensitivity to late winter precipitation compared to most other species. This may suggest that these genera may be particularly responsive to early spring moisture availability.

Maturity timing demonstrated the strongest and most consistent climate sensitivities of any phenological stage. Spring temperature variables showed uniformly strong negative correlations across all species, indicating that warmer spring conditions substantially accelerate the transition to peak canopy development. Tree of Heaven, Western Catalpa, and Common Hackberry exhibited particularly pronounced temperature sensitivity. In contrast, precipitation displayed more variable patterns across species, with some showing positive associations and others showing near-zero or slightly negative correlations. The consistency of temperature effects across all species suggests that thermal conditions are the primary driver of maturity timing, while moisture availability plays a secondary and species-specific role.

Senescence timing seems to be driven by both temperature and precipitation. Late summer maximum and mean temperature showed negative correlations across species, indicating that extreme summer heat accelerates the onset of senescence. American Lindens, Northern Red Oaks, and Siberian Elms seem to be particularly susceptible to summertime heat. In contrast, early fall temperature and late summer precipitation both showed strong positive correlations, with warmer autumn conditions and increased moisture delaying leaf color change and senescence. This suggests a dual-phase control: summer heat stress may trigger early senescence as a protective response, while favorable autumn conditions (warmth and moisture) extend the active growing season by delaying the physiological cues for leaf abscission. While temperature effects showed relatively uniform responses across species, precipitation sensitivity varied. Cottonwoods appear to be less sensitive to late-summer and early fall precipitation, possibly indicating drought tolerance.

Dormancy timing showed weaker and less consistent climate associations than earlier phenological stages. Fall precipitation metrics displayed predominantly negative correlations, suggesting that frequent autumn precipitation delays dormancy. Fall temperature showed positive but more modest correlations, indicating that warmer conditions delay dormancy. However, substantial variation in correlation strength across species existed for both temperature and precipitation variables. Siberian Elm phenology seemed to be particularly influenced by fall climate conditions, whereas Cottonwoods showed minimal climate sensitivity to fall and late fall conditions.

Comparing across all phenological stages, native tree species such as Common Hackberry and Western Catalpa, as well as invasive species Siberian Elm and Tree of Heaven, showed the highest degree of climate sensitivity, with overall absolute mean correlations exceeding. Native species Eastern and Plains Cottonwoods showed relatively low climate sensitivity across all seasons.

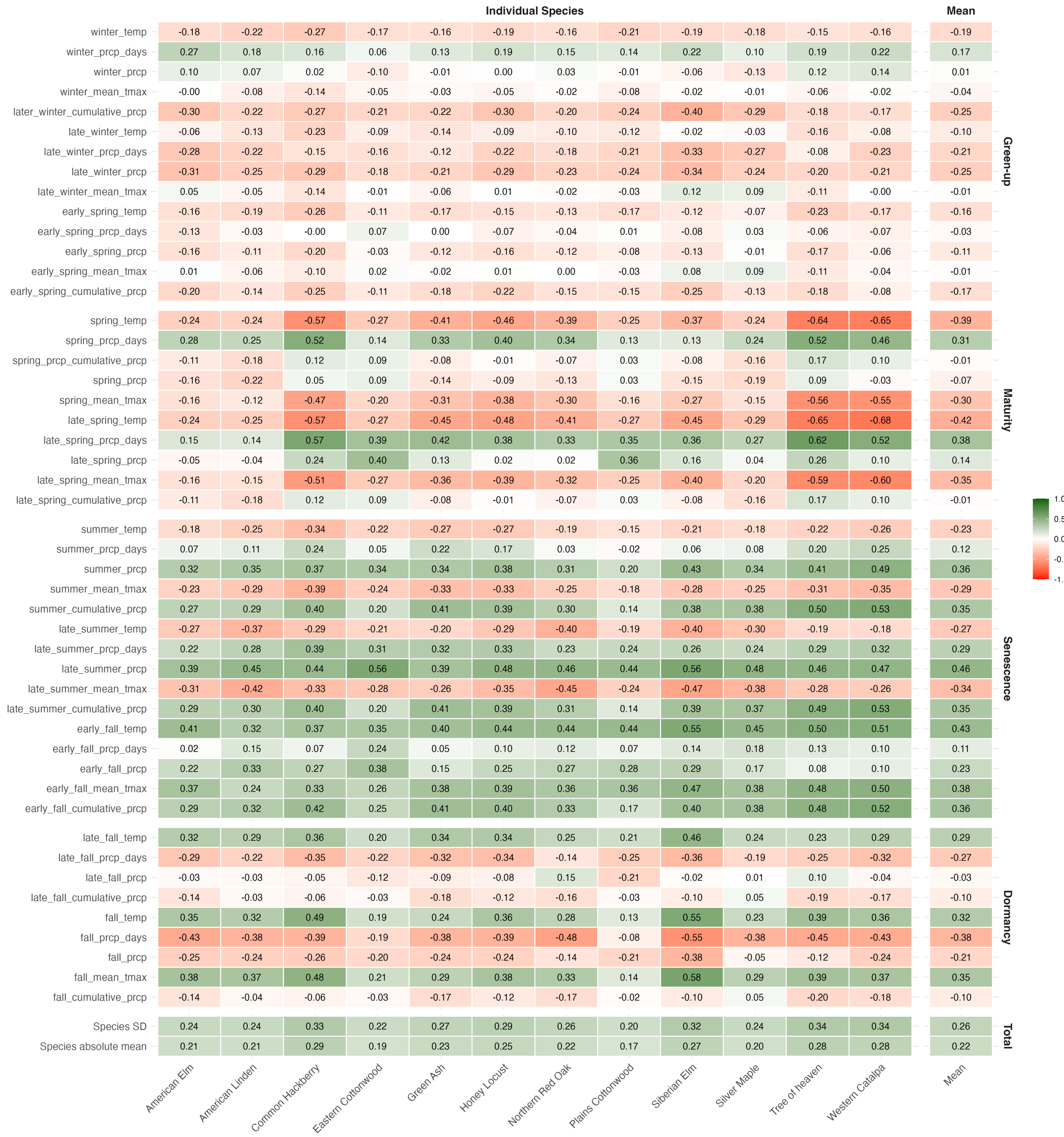

Figure 2: Pearson correlation coefficients between phenological timing and seasonal climate variables across twelve tree species. Correlations are shown for individual species and the cross-species average. Green indicates positive correlations (later phenology with higher values), red indicates negative correlations (earlier phenology with higher values). Climate variables were selected based on biological relevance to each phenological stage (see Table X for seasonal definitions).

3.1.2 Modeled Climate Responses Using Partial Dependence Analysis

While Pearson correlations provide valuable insights into bivariate climate-phenology relationships, they do not account for interactions among climate variables or control for confounding factors such as land cover and tree-level characteristics. To address these limitations, we used partial dependence analysis from the XGBoost models to examine how phenological timing responds to individual climate drivers while holding all other predictors constant at their mean values. This approach offers a more robust assessment of climate effects by isolating the marginal influence of each climate variable within the multivariate context of the full model. Partial dependence plots reveal the shape of these relationships—including potential non-linearities and thresholds (Figure 3).

Greenup responses to late winter climate displayed notable non-linear patterns. Late winter temperature effects showed a distinct optimum response across most species, with greenup occurring earliest at approximately 0°C and 3-4°C, but delayed at intermediate temperatures around 1-2°C. This U-shaped relationship contrasts with the simple negative correlation observed in bivariate analysis and suggests that both cold and warm late winter conditions can advance greenup through different physiological mechanisms. Late winter precipitation revealed a more complex relationship with greenup than suggested by correlation analysis alone. Most species exhibited an optimum response around 750mm, with greenup occurring earliest at this intermediate precipitation level and then trending toward later greenup as precipitation increased beyond this threshold. Tree of Heaven and Common Hackberry were notable exceptions, displaying relatively flat responses that plateaued or increased only slightly at the highest precipitation levels, while Eastern and Plains Cottonwoods displayed relatively flat responses throughout.

Late spring maximum temperature showed uniformly strong negative relationships with Maturity timing across all species until around 23°C, when most species exhibited a threshold response with maturity timing plateauing. The cottonwoods were notable exceptions, showing continued linear responses to temperature without evidence of plateauing, suggesting these species may maintain thermal responsiveness at higher temperatures. Spring precipitation displayed predominantly negative effects on maturity timing, with higher precipitation totals associated with earlier canopy development across most species. However, a threshold response emerged around 1800mm, beyond which most species plateaued or showed slight increases in maturity timing. Cottonwoods, Tree of Heaven, and Western Catalpa showed minimal precipitation sensitivity, with nearly flat or only gradually declining response curves.

Late summer maximum temperature showed predominantly negative relationships across all species, with higher temperatures advancing senescence timing. However, this negative impact only emerged after average maximum temperatures exceeded 32°C, beyond which most species showed a gradual decline in senescence day of year. Elms, lindens, and maples were the most affected by high temperatures, exhibiting the strongest degree of decline in senescence timing after the 32°C threshold. Cottonwoods, Northern Red Oak, Tree of Heaven, and Western Catalpa seemed to be less sensitive to extreme heat, displaying only modest declines or relatively flat response curves throughout the temperature range. Late summer precipitation

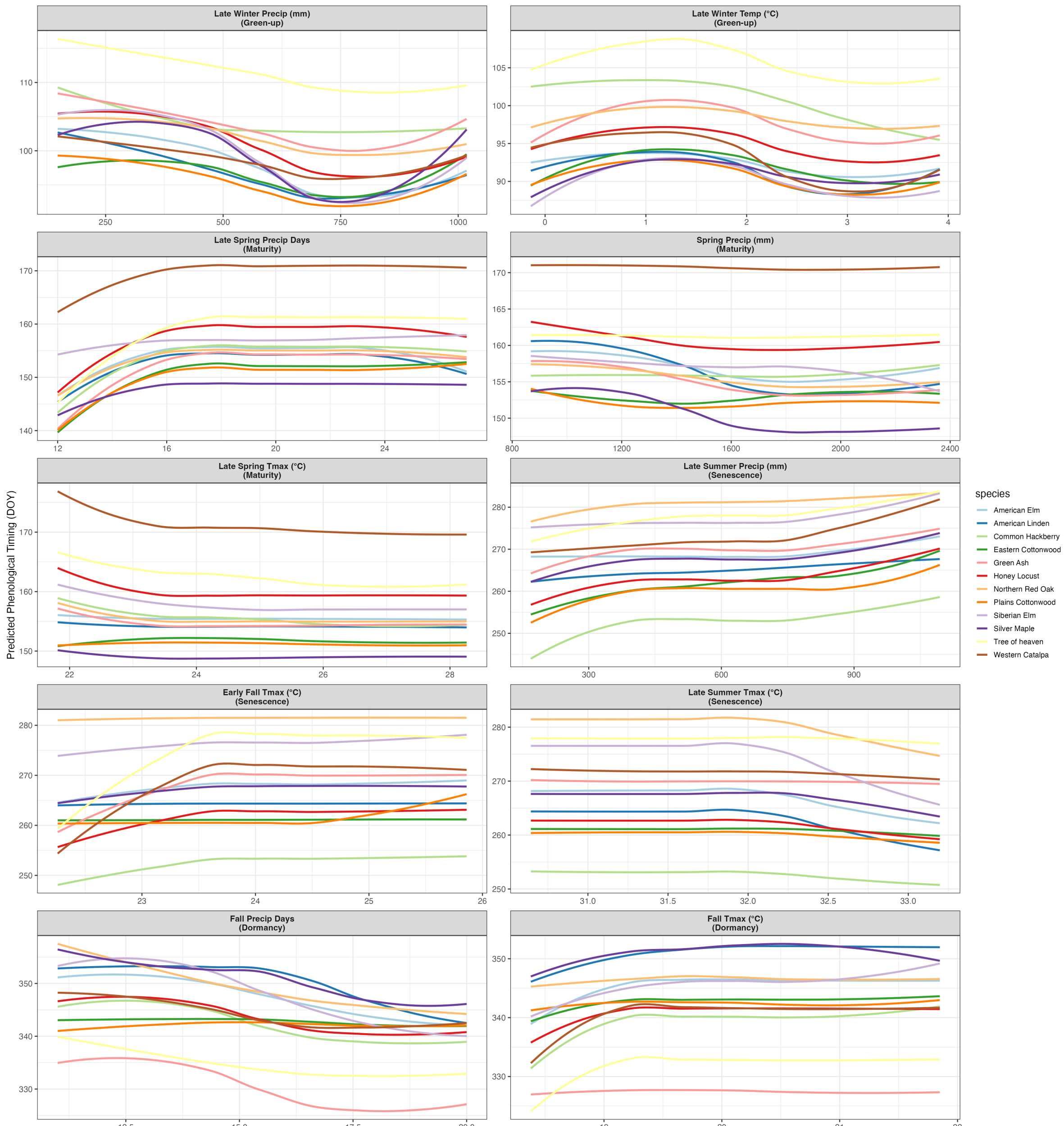

Figure 3: Partial dependence plots showing species-specific climate responses across phenological stages. Each panel shows predicted phenological timing (DOY) as a function of a key climate variable from XGBoost models, with other predictors held constant.

displayed generally positive associations with senescence timing, with higher precipitation totals delaying the onset of leaf color change. Most species in this study showed a gradual increase in senescence timing until 500mm, followed by a plateau between 500-850mm, and then a gradual increase again beyond 850mm.Early fall mean maximum temperature displayed a similar pattern of associations with senescence. Mean maximum temperature generally delayed the onset of senescence timing until it reached around 23°C, at which point senescence timing plateaued. The notable exception to this trend was Plains Cottonwood, which remained relatively flat until around 24.5°C and then senescence timing started to increase.

Fall precipitation days displayed predominantly negative relationships with Dormancy timing, with more frequent autumn precipitation days advancing complete leaf drop. Most species showed gradual linear declines, though response curve steepness varied considerably among species. Fall maximum temperature showed positive but modest effects across most species, with warmer conditions delaying dormancy. Response curves were generally shallow and variable, suggesting dormancy timing may be less tightly coupled to contemporaneous climate conditions than earlier phenological stages.

3.2 Modeling Land Cover Effects Using Partial Dependence Analysis

To examine the influence of local land cover on phenological timing, we used partial dependence analysis to isolate the effects of impervious surface and irrigated land cover while holding climate variables constant at their mean values. As impervious surface or irrigated land varied, other land cover types (tree canopy, pervious surface, etc.) were adjusted proportionally to maintain a total coverage of 100%, reflecting realistic urban land cover tradeoffs.

Partial dependence analysis revealed land cover effects that varied by phenological stage and land cover type (Figure 4). Impervious surface showed weak and inconsistent effects across phenological stages, with the overall trend suggesting greenup and maturity timing occurring later in spring and senescence occurring earlier in fall as impervious cover increased. This counterintuitive pattern—showing delayed rather than advanced spring phenology—opposes expected urban heat island effects, which would predict earlier leaf-out in warmer urban areas.

In contrast, irrigated land displayed clearer and more ecologically interpretable effects. Increased irrigation was associated with earlier greenup timing and delayed senescence across most species, consistent with enhanced water availability facilitating earlier spring development and extending the active growing season. The magnitude of irrigation effects varied considerably among species, with some showing pronounced responses (10+ day shifts across the irrigation gradient) while others, particularly the cottonwoods, showed minimal sensitivity. Maturity and dormancy stages showed weaker and more variable responses to irrigation, suggesting these transitions may be less directly coupled to current-year moisture availability.

The modest overall land cover effects compared to climate drivers suggest that while local land cover influences phenological timing, atmospheric climate conditions remain the dominant control on urban tree phenology in semi-arid environments.

3.3 Future Climate Scenario Projections

To evaluate phenological responses to climate change, we developed four warming scenarios and used our trained XGBoost models to predict phenological timing under altered climate conditions. The baseline scenario used 1970-2000 climate normals, while the +1.1°C scenario applied observed warming trends from the past 30 years uniformly across all seasons. The +2°C scenarios increased temperatures by 2°C above baseline, with three precipitation variants: moderate (no change), dry (-7% precipitation), and wet (+7% precipitation). For each scenario, we generated predictions for all species while holding land cover variables constant at their mean values, allowing us to isolate the effects of temperature and precipitation changes on phenological timing.

Climate warming scenarios revealed substantial phenological shifts across all species, with the most pronounced changes occurring in autumn phenology (Table 6, Figure 5). Under moderate warming, our models predict the overall average growing season has lengthened by 9.1 days, driven primarily by delayed dormancy rather than earlier greenup. This historical warming scenario is particularly relevant as it represents changes that have likely already occurred in Denver's urban forest.

Species responses under the +1.1°C scenario varied substantially, revealing differential adaptation trajectories. Tree of Heaven, Common Hackberry, and Western Catalpa showed the most dramatic responses, with growing seasons extending by 15-16 days through substantial dormancy delays (15-17 days later). In contrast, Northern Red Oak and cottonwoods exhibited modest growing season changes (1.5-6.6 days), maintaining relatively conservative phenological timing despite three decades of warming. Peak canopy duration under this realized warming ranged from 3.1 days (Green Ash) to 17.3 days (Siberian Elm), with most species showing 6-9 day extensions. These differential responses suggest that highly plastic species have already undergone substantial phenological shifts, while more conservative species may be operating near physiological limits or are constrained by non-climatic cues.

Notably, greenup timing remained largely unchanged under this moderate warming for most species (changes <1 day for 10 of 12 species), with only Common Hackberry and Plains Cottonwood showing slight advancement (0.7-1.9 days earlier). Maturity timing advanced modestly for most species (1-5 days earlier), while senescence delays (1-13 days) and dormancy delays (2-15 days) dominated the phenological response.

Enhanced warming scenarios (+2°C) amplified these patterns, with growing season length increasing by 10.9 days and peak canopy duration extending by 10.8 days under moderate moisture conditions. Precipitation variability modulated these responses: the +2°C wet scenario produced the largest peak canopy extensions (+14.1 days), while dry scenarios showed

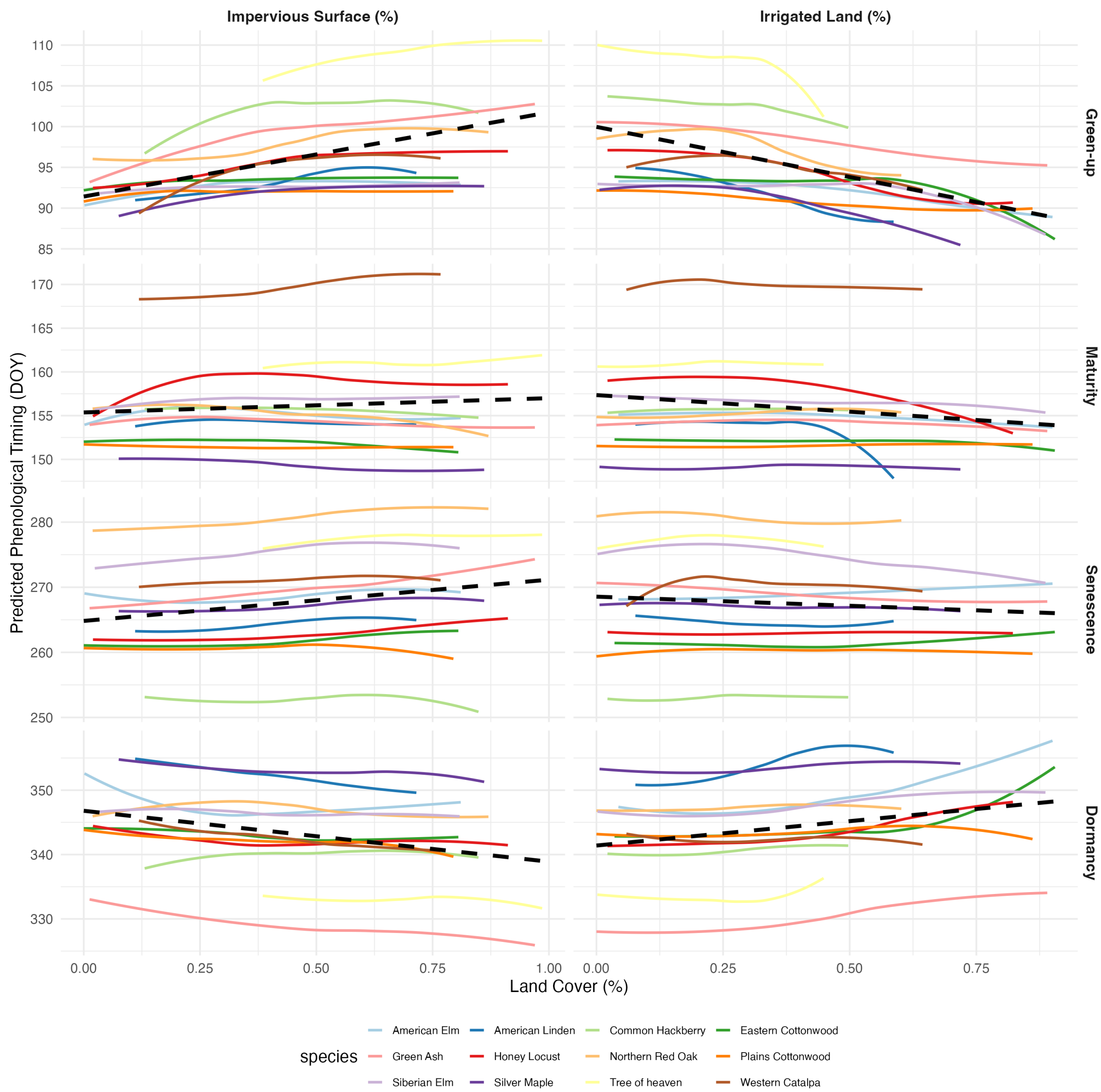

Figure 4: Partial dependence plots showing land cover effects on phenology across all stages. Each panel displays predicted phenological timing (DOY) as a function of impervious surface or irrigated land cover, with colored lines representing individual species responses and the dashed black line showing the overall linear trend across all species.

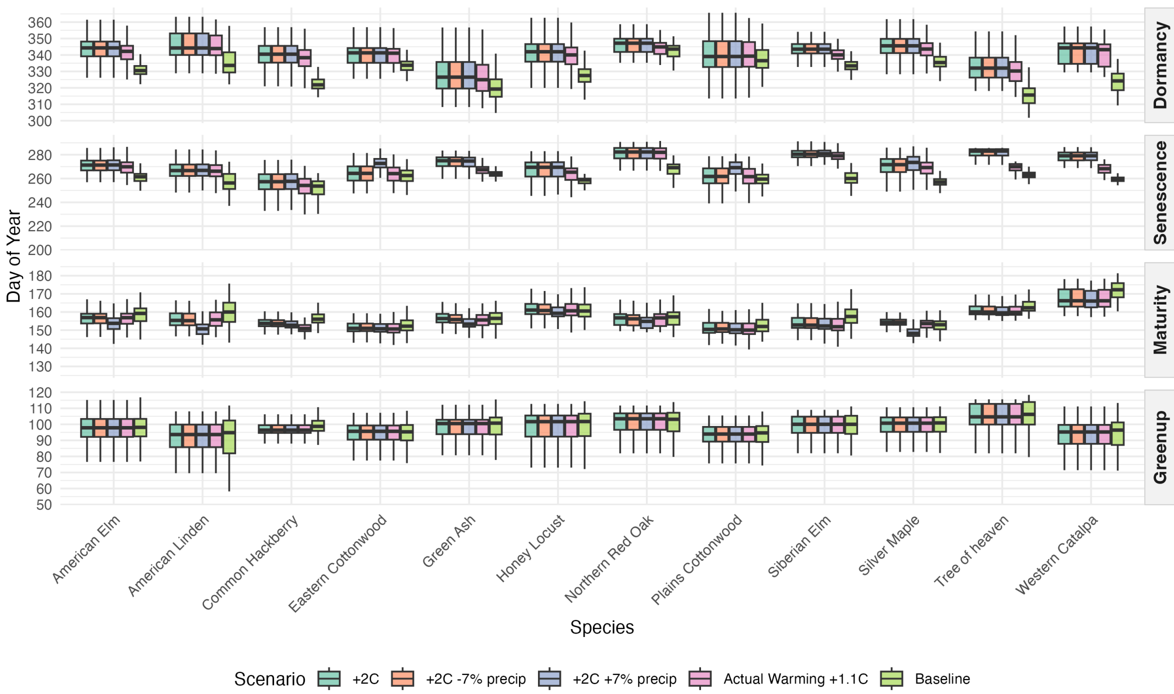

Figure 5: Phenological timing across climate warming scenarios for 12 urban tree species. Boxplots show the distribution of predicted phenological stage timing (day of year) under baseline conditions (1970-2000) and four warming scenarios. The +1.1°C scenario represents observed warming over the past 30 years, while +2°C scenarios project future warming with varying precipitation conditions (moderate, ±7% precipitation).

minimal additional effects beyond temperature alone. The progression from observed (+1.1°C) to projected (+2°C) warming reveals relatively linear responses for most phenological stages, though some species showed evidence of non-linear responses in senescence timing under enhanced warming combined with moisture changes.

The combined effect of stable greenup, modestly advanced maturity, delayed senescence, and strongly delayed dormancy suggests that the past 30 years of warming have already substantially extended both the period of active canopy cover and the total growing season length for Denver's urban forest. Continued warming is projected to amplify these patterns.

4 Discussion

4.1 Climate Sensitivity Patterns

Spring temperature showed uniformly strong negative correlations with maturity across all species, indicating warmer conditions substantially accelerate peak canopy development. In contrast, greenup and dormancy exhibited weaker and less consistent climate associations, while senescence showed intermediate sensitivity driven by both temperature and precipitation. Our findings revealed important non-linear responses and thresholds that provide insight into physiological mechanisms.

The U-shaped greenup response to late winter temperature, with earliest leaf emergence at both 0°C and 3-4°C but delays at intermediate temperatures (1-2°C), suggests that temperatures around 1-2°C are too warm to efficiently satisfy chilling requirements but too cool to accumulate sufficient heat units for rapid spring development. Late winter precipitation revealed an optimum response around 750mm, suggesting adequate moisture is necessary for cell expansion and growth. However, excessive precipitation may delay greenup by reducing soil temperatures through evaporative cooling or limiting root oxygen availability in saturated soils.

Maturity showed a threshold response at 23°C where thermal acceleration plateaus for most species, likely because photosynthetic and growth rates reach enzyme saturation limits or plants experience increasing respiration costs that offset carbon gains. Cottonwoods maintained linear responses, possibly due to their high thermal optima adapted to riparian corridors with intense summer heat. Spring precipitation displayed predominantly negative effects until around 1800mm, where adequate moisture supports rapid canopy expansion. Beyond this threshold most species plateaued, suggesting moisture is no longer limiting and other factors constrain growth.

The 32°C senescence threshold likely reflects a critical temperature where cumulative heat damage to photosynthetic machinery, increased respiration costs, and water stress from high vapor pressure deficit outweigh benefits of continued leaf retention. These temperature thresholds indicate that linear extrapolations from correlation coefficients will overestimate climate sensitivity at temperature extremes, with important implications for projecting phenological responses under future climate scenarios.

Species-specific patterns appear linked to functional traits and ecological strategies. The consistently low climate sensitivity of cottonwoods across all stages aligns with their riparian ecology and deep rooting systems, which likely buffer them from atmospheric climate variability. Species showing strong precipitation responses—Silver Maple, American Linden, and elms—are typically associated with mesic habitats where shallow-to-moderate rooting systems rely heavily on seasonal precipitation. In contrast, cottonwoods' flat precipitation curves suggest drought tolerance or access to stable groundwater sources. High sensitivity of invasive species (Tree of Heaven, Siberian Elm) may reflect phenological plasticity facilitating establishment across diverse climates. However, native species (Western Catalpa, Common Hackberry) also displayed pronounced responses, suggesting phenological plasticity represents a broader adaptive trait rather than a signature of invasiveness.

4.2 Land cover effects

The relatively modest land cover effects on phenology, particularly compared to climate drivers, challenge assumptions about strong urban heat island controls on tree phenology in semi-arid environments. The counterintuitive relationship between impervious surface and phenological timing—with increased impervious cover associated with delayed rather than advanced spring phenology—suggests that moisture limitation overwhelms thermal modification as the primary mechanism through which urbanization affects phenology in water-limited systems.

This contradiction likely stems from coupled land cover change: as impervious surface increases, irrigated areas and tree canopy necessarily decrease. In Denver's semi-arid climate, the loss of moisture resources associated with urbanization may create stress that outweighs any phenological advancement from urban warming. The stronger effects of irrigated land support this moisture-centric interpretation. Increased irrigation consistently advanced greenup and delayed senescence across most species, demonstrating that water availability directly modulates phenological timing. Further research using ground-based phenological observations paired with in-situ microclimate and soil moisture measurements would be needed to definitively separate thermal from moisture effects and validate these satellite-derived patterns

4.3 Future Climate Implications

Climate warming projections reveal divergent trajectories among urban tree species that may reshape Denver's urban forest composition. Species exhibiting high phenological plasticity—particularly, such as the Tree of Heaven, Western Catalpa, Common Hackberry, and Siberian Elm—showed substantial growing season extensions (15-23 days) and dramatic peak canopy expansions under warming scenarios. With a longer growing season, these "winner" species may gain competitive advantages through their strong responsiveness may also indicates vulnerability to climate variability. If the latter is true, invasive species (Tree of Heaven, Siberian Elm) may exploit warming conditions to expand their dominance. By contrast, cottonwoods and Northern Red Oak showed conservative responses (3-7 day growing season changes), suggesting either physiological constraints that limit climate tracking ability. These "conservative" species may face disadvantages if unable to capitalize on longer favorable conditions or their buffered responses could also provide stability under climate volatility.

For most trees extended growing seasons increase exposure periods to pests, pathogens, and late-season heat stress. The extended period of metabolic activity without corresponding increases in resource availability may induce physiological stress. Additionally, if natural enemies do not track similar climate shifts, temporal mismatches could disrupt ecosystem interactions and alter pest pressure on urban trees.

Despite these potential risks, extended growing seasons also offer important benefits for urban ecosystem services. The substantial extension of peak canopy duration (7-14 days average across scenarios) has important implications for urban ecosystem services. Longer periods of full canopy cover should enhance summer cooling benefits, with models suggesting extended shading could reduce urban heat island intensity and building energy demands.

The substantial changes already observed under +1.1°C of historical warming suggest that continued monitoring of phenological timing is particularly critical for adaptive management as Denver's Urban forest.

5. Conclusion

This study quantified species-specific phenological responses to climate drivers in Denver's urban forest using high-resolution satellite observations and machine learning methods. Climate sensitivities varied by phenological stage: spring temperature strongly controlled maturity timing, while senescence responded to both late-summer heat and early-fall precipitation. Critical temperature thresholds indicate physiological limits to phenological plasticity. Land cover effects proved modest, with increased impervious surface counterintuitively delaying spring phenology—suggesting moisture limitation overwhelms thermal modification in semiarid environments.

These climate sensitivities have already produced substantial phenological shifts. Under +1.1°C warming over the past 30 years, growing seasons have extended by 9.1 days, driven primarily by delayed dormancy rather than earlier greenup. Species responses diverged dramatically: highly plastic species extended growing seasons by 15-16 days, while conservative species showed minimal changes (3-7 days), potentially reshaping urban forest composition and favoring invasive species.

While these findings advance understanding of urban tree phenology, several limitations warrant consideration. First, sample selection and data quality constraints may introduce bias.

Only 14.5% of initially sampled trees passed quality filtering, potentially biasing results toward trees with optimal crown structure and minimal disturbance. The exclusive focus on deciduous species limits generalizability to coniferous urban forests. Additionally, temporal and spatial constraints limit inference. The seven-year study period may not capture full climate variability or multi-decadal trends, and findings specific to Denver County may not generalize to other semi-arid cities with different elevations or management practices.

Methodologically, several data limitations exist. Satellite-derived phenological dates lack ground-truth validation against field observations of actual leaf emergence or senescence. Land cover was treated as static despite likely changes during 2018-2024, potentially masking urbanization effects, and the "irrigated land" classification does not capture irrigation intensity or scheduling variability. Weather station data may not represent fine-scale microclimate variation across diverse neighborhoods and topography.

Furthermore, remote sensing cannot account for important biological complexity. The approach cannot capture fine-scale physiological stress indicators, distinguish between phenological advancement and stress-induced leaf loss, or account for individual tree variation due to age, genetics, health status, or management history (pruning, fertilization, supplemental watering).

Finally, our modeling framework relies on simplifying assumptions. Models assume climate-phenology relationships remain temporally stable and apply simplified future climate scenarios with uniform seasonal warming, though actual climate change involves complex seasonal shifts, increased variability, and potential shifts in tree sensitivity as they acclimate or reach physiological limits. Ground-based validation with in-situ microclimate and soil moisture measurements would strengthen causal inference, particularly for disentangling thermal from moisture mechanisms.

Despite these limitations, this study demonstrates that climate change has substantially altered urban tree phenology in Denver. Extended growing seasons should enhance cooling benefits but also increase risks from physiological stress and pest exposure. Effective climate adaptation requires species selection strategies that balance phenological plasticity with stability and active monitoring to detect early warning signals of climate stress.

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7. Appendix

Table 5: XGBoost model performance by species and phenological stage

phenology	species	Total_Obs	Mean_R2	Mean_RMSE	Mean_MAE
Greenup_doy	American Elm	1	0.658	8.421	6.514
Greenup_doy	American Linden	1	0.700	8.703	6.724
Greenup_doy	Common Hackberry	1	0.571	8.481	6.579
Greenup_doy	Eastern Cottonwood	1	0.628	7.318	5.496
Greenup_doy	Green Ash	1	0.679	8.130	6.099
Greenup_doy	Honey Locust	1	0.632	8.785	6.687
Greenup_doy	Northern Red Oak	1	0.556	9.042	6.852
Greenup_doy	Plains Cottonwood	1	0.761	6.839	5.146
Greenup_doy	Siberian Elm	1	0.642	7.860	6.026
Greenup_doy	Silver Maple	1	0.648	7.075	5.368
Greenup_doy	Tree of heaven	1	0.691	8.365	6.356
Greenup_doy	Western Catalpa	1	0.558	10.480	8.004
Maturity_doy	American Elm	1	0.601	6.718	5.093
Maturity_doy	American Linden	1	0.611	7.432	5.244
Maturity_doy	Common Hackberry	1	0.719	6.622	4.897
Maturity_doy	Eastern Cottonwood	1	0.621	5.903	4.216
Maturity_doy	Green Ash	1	0.700	6.694	4.968
Maturity_doy	Honey Locust	1	0.688	7.385	5.465
Maturity_doy	Northern Red Oak	1	0.591	7.176	5.122
Maturity_doy	Plains Cottonwood	1	0.563	6.655	4.754
Maturity_doy	Siberian Elm	1	0.607	7.745	5.849
Maturity_doy	Silver Maple	1	0.604	6.459	4.767

Table 5: XGBoost model performance by species and phenological stage

phenology	species	Total_Obs	Mean_R2	Mean_RMSE	Mean_MAE
Maturity_doy	Tree of heaven	1	0.721	6.539	4.657
Maturity_doy	Western Catalpa	1	0.696	8.067	6.379
Senescence_doy	American Elm	1	0.680	8.070	6.308
Senescence_doy	American Linden	1	0.637	8.540	6.709
Senescence_doy	Common Hackberry	1	0.685	10.109	7.422
Senescence_doy	Eastern Cottonwood	1	0.737	9.207	6.818
Senescence_doy	Green Ash	1	0.661	8.681	6.649
Senescence_doy	Honey Locust	1	0.705	8.387	6.578
Senescence_doy	Northern Red Oak	1	0.607	8.998	6.882
Senescence_doy	Plains Cottonwood	1	0.777	9.197	7.308
Senescence_doy	Siberian Elm	1	0.672	10.846	8.391
Senescence_doy	Silver Maple	1	0.712	8.896	6.864
Senescence_doy	Tree of heaven	1	0.696	9.449	7.176
Senescence_doy	Western Catalpa	1	0.720	8.969	7.065
Dormancy_doy	American Elm	1	0.628	11.994	9.138
Dormancy_doy	American Linden	1	0.649	11.603	9.245
Dormancy_doy	Common Hackberry	1	0.606	10.996	8.555
Dormancy_doy	Eastern Cottonwood	1	0.513	11.896	8.469
Dormancy_doy	Green Ash	1	0.633	12.899	9.959
Dormancy_doy	Honey Locust	1	0.591	11.317	8.694
Dormancy_doy	Northern Red Oak	1	0.533	11.465	8.797
Dormancy_doy	Plains Cottonwood	1	0.621	11.729	8.927
Dormancy_doy	Siberian Elm	1	0.587	11.405	8.425
Dormancy_doy	Silver Maple	1	0.523	11.647	8.902
Dormancy_doy	Tree of heaven	1	0.612	11.189	8.731
Dormancy_doy	Western Catalpa	1	0.580	11.594	8.624

phenology	species	tree_mean_pheno	late_winter_prcp	impervious_90m	late_winter_temp	late_winter_prcp_days	irrigated_90m	UID_numeric	tree_canopy_90m	late_winter_mean_tmax	tree_n_obs	unirrigated_90m	water_90m	late_spring_prcp_days	late_spring_temp	spring_prcp	late_spring_mean_tmax	early_fall_mean_tmax	late_summer_mean_tmax	late_summer_prcp	early_fall_temp	late_summer_temp	late_summer_prcp_days	fall_prcp_days	fall_mean_tmax	fall_cumulative_prcp	late_fall_temp
Greenup_doy	**American Elm**	0.604	0.162	0.068	0.043	0.038	0.036	0.017	0.015	0.009	0.003	0.002	0.002	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	American Linden	0.564	0.068	0.071	0.086	0.029	0.109	0.026	0.024	0.011	0.007	0.003	0.003	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Common Hackberry	0.6	0.08	0.07	0.088	0.029	0.043	0.023	0.026	0.023	0.005	0.01	0.003	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Eastern Cottonwood	0.616	0.096	0.032	0.094	0.019	0.071	0.013	0.029	0.014	0.003	0.009	0.004	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Green Ash	0.679	0.073	0.069	0.087	0.018	0.037	0.006	0.011	0.013	0.002	0.002	0.002	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Honey Locust	0.671	0.112	0.048	0.064	0.029	0.025	0.012	0.014	0.011	0.006	0.006	0.002	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Northern Red Oak	0.56	0.079	0.089	0.085	0.028	0.047	0.032	0.046	0.016	0.01	0.002	0.006	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Plains Cottonwood	0.654	0.081	0.018	0.093	0.017	0.08	0.009	0.009	0.012	0.004	0.019	0.004	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Siberian Elm	0.519	0.244	0.023	0.092	0.042	0.016	0.014	0.023	0.018	0.003	0.003	0.003	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Silver Maple	0.524	0.283	0.011	0.113	0.021	0.016	0.006	0.007	0.015	0.002	0.001	0.001	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Tree of heaven	0.714	0.054	0.038	0.057	0.018	0.037	0.022	0.022	0.03	0.003	0.002	0.003	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Western Catalpa	0.509	0.068	0.09	0.1	0.047	0.07	0.052	0.033	0.02	0.012	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Maturity_doy	**American Elm**	0.52	-	0.03	-	-	0.03	0.016	0.025	-	0.003	0.002	0.004	0.212	0.072	0.066	0.02	-	-	-	-	-	-	-	-	-	-
	American Linden	0.422	-	0.048	-	-	0.043	0.025	0.037	-	0.006	0.006	0.007	0.176	0.085	0.127	0.021	-	-	-	-	-	-	-	-	-	-
	Common Hackberry	0.185	-	0.022	-	-	0.016	0.021	0.027	-	0.001	0.003	0.003	0.41	0.229	0.018	0.066	-	-	-	-	-	-	-	-	-	-
	Eastern Cottonwood	0.291	-	0.053	-	-	0.018	0.01	0.023	-	0.003	0.024	0.004	0.451	0.052	0.05	0.02	-	-	-	-	-	-	-	-	-	-
	Green Ash	0.233	-	0.021	-	-	0.015	0.01	0.017	-	0.003	0.001	0.002	0.356	0.185	0.11	0.048	-	-	-	-	-	-	-	-	-	-
	Honey Locust	0.319	-	0.02	-	-	0.024	0.009	0.017	-	0.003	0.004	0.002	0.201	0.243	0.032	0.125	-	-	-	-	-	-	-	-	-	-
	Northern Red Oak	0.331	-	0.039	-	-	0.034	0.037	0.086	-	0.003	0.001	0.004	0.154	0.2	0.057	0.052	-	-	-	-	-	-	-	-	-	-
	Plains Cottonwood	0.396	-	0.033	-	-	0.034	0.016	0.037	-	0.002	0.033	0.021	0.302	0.068	0.041	0.017	-	-	-	-	-	-	-	-	-	-
	Siberian Elm	0.47	-	0.022	-	-	0.03	0.016	0.033	-	0.006	0.002	0.002	0.056	0.247	0.03	0.086	-	-	-	-	-	-	-	-	-	-
	Silver Maple	0.404	-	0.018	-	-	0.015	0.007	0.014	-	0.002	0.001	0.001	0.268	0.128	0.117	0.026	-	-	-	-	-	-	-	-	-	-
	Tree of heaven	0.21	-	0.026	-	-	0.033	0.029	0.032	-	0.004	0.001	0.002	0.323	0.226	0.015	0.1	-	-	-	-	-	-	-	-	-	-
	Western Catalpa	0.263	-	0.037	-	-	0.043	0.027	0.026	-	0.014	-	-	0.112	0.346	0.03	0.102	-	-	-	-	-	-	-	-	-	-
Senescence_doy	**American Elm**	0.504	-	0.059	-	-	0.03	0.009	0.046	-	0.004	0.002	0.003	-	-	-	-	0.083	0.077	0.066	0.064	0.031	0.022	-	-	-	-
	American Linden	0.525	-	0.053	-	-	0.03	0.022	0.035	-	0.008	0.003	0.001	-	-	-	-	0.012	0.096	0.086	0.065	0.037	0.027	-	-	-	-
	Common Hackberry	0.488	-	0.038	-	-	0.021	0.022	0.033	-	0.005	0.004	0.002	-	-	-	-	0.081	0.026	0.153	0.038	0.017	0.072	-	-	-	-
	Eastern Cottonwood	0.421	-	0.027	-	-	0.026	0.006	0.021	-	0.004	0.032	0.003	-	-	-	-	0.014	0.022	0.201	0.073	0.043	0.109	-	-	-	-
	Green Ash	0.478	-	0.059	-	-	0.024	0.008	0.017	-	0.002	0.002	0.002	-	-	-	-	0.156	0.007	0.107	0.063	0.006	0.071	-	-	-	-
	Honey Locust	0.489	-	0.025	-	-	0.016	0.011	0.025	-	0.002	0.002	0.001	-	-	-	-	0.093	0.044	0.156	0.069	0.011	0.053	-	-	-	-
	Northern Red Oak	0.451	-	0.052	-	-	0.03	0.04	0.033	-	0.007	0.001	0.014	-	-	-	-	0.015	0.084	0.072	0.112	0.038	0.051	-	-	-	-
	Plains Cottonwood	0.42	-	0.015	-	-	0.039	0.005	0.011	-	0.001	0.019	0.015	-	-	-	-	0.066	0.015	0.139	0.047	0.059	0.149	-	-	-	-
	Siberian Elm	0.202	-	0.021	-	-	0.018	0.007	0.024	-	0.007	0.003	0.003	-	-	-	-	0.03	0.176	0.07	0.348	0.072	0.017	-	-	-	-
	Silver Maple	0.47	-	0.008	-	-	0.007	0.002	0.012	-	0.001	0	0	-	-	-	-	0.023	0.091	0.091	0.263	0	0.031	-	-	-	-
	Tree of heaven	0.362	-	0.035	-	-	0.06	0.023	0.055	-	0.019	0.006	0.004	-	-	-	-	0.154	0.008	0.146	0.061	0.003	0.064	-	-	-	-
	Western Catalpa	0.375	-	0.039	-	-	0.052	0.035	0.026	-	0.01	-	-	-	-	-	-	0.205	0.013	0.142	0.061	0.004	0.038	-	-	-	-
Dormancy_doy	**American Elm**	0.465	-	0.05	-	-	0.034	0.013	0.043	-	0.002	0.003	0.003	-	-	-	-	-	-	-	-	-	-	0.122	0.094	0.087	0.085
	American Linden	0.514	-	0.039	-	-	0.065	0.031	0.031	-	0.004	0.005	0.007	-	-	-	-	-	-	-	-	-	-	0.126	0.057	0.057	0.065
	Common Hackberry	0.492	-	0.04	-	-	0.019	0.014	0.033	-	0.003	0.018	0.003	-	-	-	-	-	-	-	-	-	-	0.14	0.164	0.02	0.056
	Eastern Cottonwood	0.508	-	0.036	-	-	0.098	0.015	0.037	-	0.005	0.024	0.013	-	-	-	-	-	-	-	-	-	-	0.064	0.097	0.031	0.071
	Green Ash	0.611	-	0.032	-	-	0.023	0.011	0.013	-	0.002	0.002	0.002	-	-	-	-	-	-	-	-	-	-	0.094	0.025	0.085	0.099
	Honey Locust	0.529	-	0.029	-	-	0.024	0.012	0.025	-	0.003	0.006	0.001	-	-	-	-	-	-	-	-	-	-	0.127	0.118	0.034	0.093
	Northern Red Oak	0.434	-	0.055	-	-	0.036	0.032	0.032	-	0.012	0.002	0.012	-	-	-	-	-	-	-	-	-	-	0.179	0.048	0.099	0.057
	Plains Cottonwood	0.596	-	0.029	-	-	0.052	0.016	0.027	-	0.003	0.045	0.011	-	-	-	-	-	-	-	-	-	-	0.053	0.04	0.059	0.069
	Siberian Elm	0.371	-	0.031	-	-	0.022	0.024	0.057	-	0.008	0.006	0.008	-	-	-	-	-	-	-	-	-	-	0.231	0.099	0.023	0.12
	Silver Maple	0.532	-	0.018	-	-	0.028	0.011	0.033	-	0.003	0.001	0.001	-	-	-	-	-	-	-	-	-	-	0.158	0.128	0.056	0.032
	Tree of heaven	0.464	-	0.026	-	-	0.043	0.026	0.02	-	0.005	0.006	0.002	-	-	-	-	-	-	-	-	-	-	0.137	0.162	0.052	0.057
	Western Catalpa	0.413	-	0.045	-	-	0.05	0.03	0.044	-	0.011	-	-	-	-	-	-	-	-	-	-	-	-	0.115	0.153	0.089	0.049

Table 6: Phenological timing (day of year) and changes (Δ) for 12 tree species under climate warming scenarios

Species	Phenology	Baseline	+1.1°C Mean	+1.1°C Δ	+1.1°C p	+2°C Mean	+2°C Δ	+2°C p	+2°C Dry Mean	+2°C Dry Δ	+2°C Dry p	+2°C Wet Mean	+2°C Wet Δ	+2°C Wet p
American Elm	Greenup	97.0	97.0	0.0	1.000	97.0	0.0	1.000	97.0	0.0	1.000	97.0	0.0	1.000
	Maturity	158.4	156.3	-2.1	<0.001	156.3	-2.2	<0.001	156.3	-2.1	<0.001	153.7	-4.8	<0.001
	Senescence	263.3	267.9	4.7	<0.001	271.5	8.2	<0.001	271.5	8.2	<0.001	271.9	8.7	<0.001
	Dormancy	332.0	341.7	9.7	<0.001	344.0	12.0	<0.001	344.0	12.0	<0.001	344.0	12.0	<0.001
	Growing Season	*235.0*	*244.7*	*9.7*		*247.0*	*12.0*		*247.0*	*12.0*		*247.0*	*12.0*	
	Peak Canopy	*104.8*	*111.6*	*6.8*		*115.2*	*10.4*		*115.2*	*10.4*		*118.3*	*13.5*	
American Linden	Greenup	91.1	91.6	0.5	0.300	91.6	0.5	0.300	91.6	0.5	0.300	91.6	0.5	0.300
	Maturity	160.4	156.3	-4.2	<0.001	156.2	-4.3	<0.001	156.1	-4.4	<0.001	150.9	-9.5	<0.001
	Senescence	262.3	266.1	3.8	<0.001	266.3	4.1	<0.001	266.3	4.1	<0.001	266.4	4.1	<0.001
	Dormancy	335.9	344.4	8.5	<0.001	345.1	9.2	<0.001	345.1	9.2	<0.001	345.1	9.2	<0.001
	Growing Season	*244.9*	*252.9*	*8.0*		*253.6*	*8.7*		*253.6*	*8.7*		*253.6*	*8.7*	
	Peak Canopy	*101.8*	*109.8*	*8.0*		*110.2*	*8.4*		*110.3*	*8.4*		*115.4*	*13.6*	
Common Hackberry	Greenup	98.0	96.1	-1.9	<0.001	96.1	-1.9	<0.001	96.1	-1.9	<0.001	96.1	-1.9	<0.001
	Maturity	156.4	151.2	-5.2	<0.001	154.0	-2.4	<0.001	153.9	-2.5	<0.001	153.2	-3.2	<0.001
	Senescence	251.1	252.4	1.3	<0.001	257.3	6.2	<0.001	257.3	6.2	<0.001	257.5	6.4	<0.001
	Dormancy	323.4	338.5	15.1	<0.001	340.8	17.3	<0.001	340.8	17.3	<0.001	340.8	17.3	<0.001
	Growing Season	*225.5*	*242.4*	*16.9*		*244.6*	*19.2*		*244.6*	*19.2*		*244.6*	*19.2*	
	Peak Canopy	*94.7*	*101.2*	*6.5*		*103.3*	*8.7*		*103.5*	*8.8*		*104.3*	*9.6*	
Eastern Cottonwood	Greenup	94.3	94.4	0.1	0.300	94.4	0.1	0.300	94.4	0.1	0.300	94.4	0.1	0.300
	Maturity	152.5	150.9	-1.6	<0.001	151.3	-1.3	<0.001	151.3	-1.2	<0.001	151.0	-1.5	<0.001
	Senescence	262.7	264.5	1.9	<0.001	265.0	2.3	<0.001	265.0	2.3	<0.001	272.6	9.9	<0.001
	Dormancy	334.2	340.9	6.6	<0.001	341.2	6.9	<0.001	341.2	6.9	<0.001	341.2	6.9	<0.001
	Growing Season	*239.9*	*246.5*	*6.6*		*246.8*	*6.9*		*246.8*	*6.9*		*246.8*	*6.9*	
	Peak Canopy	*110.2*	*113.7*	*3.5*		*113.8*	*3.6*		*113.7*	*3.5*		*121.6*	*11.4*	
Green Ash	Greenup	98.2	98.0	-0.2	<0.001	98.0	-0.2	<0.001	98.0	-0.2	<0.001	98.0	-0.2	<0.001
	Maturity	156.4	155.4	-0.9	<0.001	156.5	0.2	<0.001	156.1	-0.3	<0.001	153.8	-2.5	<0.001
	Senescence	263.8	266.0	2.2	<0.001	273.9	10.0	<0.001	273.9	10.0	<0.001	273.7	9.9	<0.001
	Dormancy	320.4	326.3	5.9	<0.001	327.7	7.3	<0.001	327.7	7.3	<0.001	327.7	7.3	<0.001
	Growing Season	*222.2*	*228.3*	*6.1*		*229.7*	*7.5*		*229.7*	*7.5*		*229.7*	*7.5*	
	Peak Canopy	*107.4*	*110.6*	*3.1*		*117.3*	*9.9*		*117.8*	*10.3*		*119.9*	*12.5*	
Honey Locust	Greenup	99.0	98.6	-0.3	<0.001	98.6	-0.3	<0.001	98.6	-0.3	<0.001	98.6	-0.3	<0.001
	Maturity	160.7	161.2	0.5	<0.001	161.5	0.8	<0.001	161.3	0.6	<0.001	160.1	-0.6	<0.001
	Senescence	259.4	263.0	3.6	<0.001	267.4	8.0	<0.001	267.4	8.0	<0.001	267.6	8.2	<0.001
	Dormancy	327.8	339.4	11.6	<0.001	341.1	13.3	<0.001	341.1	13.3	<0.001	341.1	13.3	<0.001
	Growing Season	*228.8*	*240.8*	*11.9*		*242.5*	*13.6*		*242.5*	*13.6*		*242.5*	*13.6*	
	Peak Canopy	*98.7*	*101.8*	*3.2*		*105.9*	*7.2*		*106.1*	*7.4*		*107.4*	*8.8*	
Northern Red Oak	Greenup	100.3	101.0	0.7	<0.001	101.0	0.7	<0.001	101.0	0.7	<0.001	101.0	0.7	<0.001
	Maturity	156.8	156.4	-0.4	<0.001	156.6	-0.2	0.100	156.2	-0.7	<0.001	154.9	-1.9	<0.001
	Senescence	272.8	281.4	8.6	<0.001	281.6	8.8	<0.001	281.6	8.8	<0.001	281.7	8.9	<0.001
	Dormancy	342.7	344.9	2.2	<0.001	347.0	4.3	<0.001	347.0	4.3	<0.001	347.0	4.3	<0.001
	Growing Season	*242.4*	*243.9*	*1.5*		*245.9*	*3.5*		*245.9*	*3.5*		*245.9*	*3.5*	
	Peak Canopy	*116.0*	*125.1*	*9.1*		*125.0*	*9.0*		*125.5*	*9.5*		*126.8*	*10.8*	
Plains Cottonwood	Greenup	92.4	91.8	-0.7	<0.001	91.8	-0.7	<0.001	91.8	-0.7	<0.001	91.8	-0.7	<0.001
	Maturity	152.7	150.7	-2.0	<0.001	151.1	-1.5	<0.001	151.4	-1.2	<0.001	150.8	-1.8	<0.001
	Senescence	259.3	261.1	1.8	<0.001	261.4	2.0	<0.001	261.4	2.0	<0.001	268.0	8.7	<0.001
	Dormancy	337.9	340.4	2.5	<0.001	340.7	2.8	<0.001	340.7	2.8	<0.001	340.7	2.8	<0.001
	Growing Season	*245.5*	*248.7*	*3.2*		*249.0*	*3.5*		*249.0*	*3.5*		*249.0*	*3.5*	
	Peak Canopy	*106.7*	*110.4*	*3.8*		*110.2*	*3.6*		*109.9*	*3.3*		*117.2*	*10.5*	
Siberian Elm	Greenup	98.8	98.7	-0.1	0.100	98.7	-0.1	0.100	98.7	-0.1	0.100	98.7	-0.1	0.100
	Maturity	158.0	153.2	-4.8	<0.001	154.0	-4.0	<0.001	153.9	-4.2	<0.001	153.6	-4.5	<0.001
	Senescence	265.3	277.8	12.5	<0.001	279.9	14.5	<0.001	279.9	14.5	<0.001	281.1	15.8	<0.001
	Dormancy	333.7	340.3	6.6	<0.001	343.9	10.2	<0.001	343.9	10.2	<0.001	343.9	10.2	<0.001
	Growing Season	*234.9*	*241.6*	*6.7*		*245.2*	*10.3*		*245.2*	*10.3*		*245.2*	*10.3*	
	Peak Canopy	*107.3*	*124.6*	*17.3*		*125.8*	*18.5*		*126.0*	*18.7*		*127.6*	*20.3*	
Silver Maple	Greenup	99.5	99.4	-0.1	<0.001	99.4	-0.1	<0.001	99.4	-0.1	<0.001	99.4	-0.1	<0.001
	Maturity	152.9	153.4	0.5	<0.001	154.6	1.7	<0.001	154.5	1.6	<0.001	149.0	-3.9	<0.001
	Senescence	259.7	268.5	8.9	<0.001	270.1	10.4	<0.001	270.1	10.4	<0.001	271.2	11.5	<0.001
	Dormancy	335.8	343.3	7.5	<0.001	345.4	9.6	<0.001	345.4	9.6	<0.001	345.4	9.6	<0.001
	Growing Season	*236.3*	*243.9*	*7.6*		*246.0*	*9.7*		*246.0*	*9.7*		*246.0*	*9.7*	
	Peak Canopy	*106.8*	*115.2*	*8.4*		*115.5*	*8.7*		*115.6*	*8.8*		*122.2*	*15.4*	
Tree of heaven	Greenup	104.6	104.2	-0.4	0.100	104.2	-0.4	0.100	104.2	-0.4	0.100	104.2	-0.4	0.100
	Maturity	163.4	160.9	-2.5	<0.001	161.2	-2.2	<0.001	161.1	-2.3	<0.001	160.6	-2.8	<0.001
	Senescence	263.7	269.1	5.4	<0.001	280.6	16.9	<0.001	280.6	16.9	<0.001	280.7	17.0	<0.001
	Dormancy	315.0	330.2	15.2	<0.001	332.8	17.7	<0.001	332.8	17.7	<0.001	332.8	17.7	<0.001
	Growing Season	*210.4*	*226.0*	*15.6*		*228.5*	*18.1*		*228.5*	*18.1*		*228.5*	*18.1*	
	Peak Canopy	*100.3*	*108.2*	*7.9*		*119.4*	*19.2*		*119.5*	*19.2*		*120.0*	*19.7*	
Western Catalpa	Greenup	92.3	93.2	1.0	0.100	93.2	1.0	0.100	93.2	1.0	0.100	93.2	1.0	0.100
	Maturity	172.1	167.7	-4.4	<0.001	167.8	-4.3	<0.001	167.8	-4.3	<0.001	167.3	-4.8	<0.001
	Senescence	259.5	266.4	7.0	<0.001	278.0	18.6	<0.001	278.0	18.6	<0.001	278.0	18.6	<0.001
	Dormancy	324.4	341.3	16.9	<0.001	342.7	18.3	<0.001	342.7	18.3	<0.001	342.7	18.3	<0.001
	Growing Season	*232.1*	*248.1*	*16.0*		*249.5*	*17.4*		*249.5*	*17.4*		*249.5*	*17.4*	
	Peak Canopy	*87.3*	*98.8*	*11.4*		*110.3*	*22.9*		*110.2*	*22.9*		*110.7*	*23.4*	
Overall Average	***Growing Season***	***233.2***	***242.3***	***9.1***		***244.0***	***10.9***		***244.0***	***10.9***		***244.0***	***10.9***	
Overall Average	***Peak Canopy***	***103.5***	***110.9***	***7.4***		***114.3***	***10.8***		***114.4***	***10.9***		***117.6***	***14.1***	